List and Map Variables

To Begin with the Lab

Summary of the Lab

This lab introduces Terraform composite variables, specifically list and map variables. List variables are used to store multiple values, such as security groups, while map variables store key-value pairs, such as AWS regions and their corresponding AMI IDs. By using these variable types, Terraform configurations become more flexible, reusable, and environment-independent, allowing deployments to adapt easily to different requirements without modifying the core infrastructure code.

- Steps
  - Open variables.tf and create a list variable named security_group.
  - Define the variable type as list and add multiple security group IDs as default values.
  - Open create-instance.tf and attach the list variable to the EC2 instance's security_groups attribute.
  - Save the files, commit the changes, and pull the latest code on the Terraform machine.
- Run:

```
terraform plan
```

- To verify that the security group list is included in the execution plan.
- In variables.tf, create a map variable named amis.
- Define the variable type as map and add AWS region-to-AMI ID key-value pairs.
- Replace the hardcoded AMI in create-instance.tf with a lookup() function that retrieves the AMI based on the selected region.

```
variable "Security_Group"{
    type = list
    default = ["sg-24076", "sg-90890", "sg-456789"]
}

variable "AMIS" {
    type = map
    default = {
        us-east-1 = "ami-0b6d9d3d33ba97d99"
        us-east-2 = "ami-05692172625678b4e"
    }
}
```

- Save the changes, commit the code, and pull the updates on the Terraform machine.
- Run:

```
terraform plan
```

- To verify that the AMI is selected from the map variable.

- Execute Terraform again with a different region value and confirm that the corresponding AMI changes automatically.
- Review the plan output to validate both list and map variable functionality.